Masruk Habib

Computer Engineer · R&D Engineer (Full Stack, LLMs and RAG) · Data Analyst · Researcher

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ABOUT ME

I build AI systems that make it into production. At GoScurry.ai I work across the stack on an AI meeting intelligence platform, from the React onboarding flow through the FastAPI services to the RAG layer and the CRM integrations behind them. Before this I spent eight months at Jatri as a data analyst, answering sales, finance and supply chain questions for a business that needed them answered quickly. Five of my papers are published, most of them on applying machine learning to health problems, and for a year and a half I showed other students how to get their own work into print.

EXPERIENCE

Research & Development Engineer · GoScurry.ai — **Apr 2026 to present**

Dhaka, Bangladesh · AI meeting intelligence platform, React 19 and TypeScript on FastAPI and PostgreSQL

- Second largest contributor on a six person team: 366 merged pull requests and roughly 140,000 lines of production code across frontend, backend, database schema and CI, alongside 223 test files and 49 Alembic migrations.
- Own customer onboarding end to end through two full rewrites, covering 22 backend modules and an eight screen React flow that composes, validates and atomically commits an executable workflow graph for each account.
- Built the RAG personalisation layer on pgvector with HNSW indexing, retrieval logging and tuning, and a Batch API pipeline that keeps bulk embedding affordable.
- Designed the Pipedrive CRM integration: AI ranked field shortlisting that narrows 49 fields to 10, field descriptions generated from real data, mapping profiles, and a per field human review gate before anything is written back.
- Authored the platform's prompt injection defence, applied to every untrusted transcript, email body and fetched page, and shipped the billing and growth systems behind spend caps, usage metering, low balance alerts and referral commissions.

Data Analyst · Jatri — **Sep 2025 to Apr 2026**

Gulshan, Dhaka, Bangladesh

- Dug into sales, marketing, finance and supply chain data to answer the questions the business actually had.
- Built the dashboards and automated the reporting, so numbers arrived without anyone chasing them.
- Tracked sales growth, inventory and market trends, and watched how customers responded to promotions.
- Helped with demand forecasting and sales planning, flagging risks and openings as they appeared.
- Kept data consistent across systems and worked with whichever team needed numbers that week.

Backend Developer · CRM Squirrel — **Aug 2025 to Sep 2025**

Lampang, Thailand (remote)

- Wrote Python backend services behind core CRM workflows.
- Handled API integration and helped shape how the system would scale.

Student Research Mentor · Dept. of Computer Engineering, Marwadi University — **Nov 2023 to Apr 2025**

Rajkot, Gujarat, India

- Took three batches of students from first draft through to a published paper.
- Ran 37 sessions at the Research Activity Club on machine learning and on writing results up properly.
- Co-authored two papers with students from other countries.

Research Intern, SRIT · Dept. of Computer Engineering, Marwadi University — **May 2024 to Jun 2024**

Rajkot, Gujarat, India

- Wrote and published one paper over the two months, under a faculty mentor.

EDUCATION

B.Tech in Computer Engineering — **Nov 2021 to Jun 2025**

Marwadi University, Rajkot, Gujarat, India · CGPA 7.64 / 10.00, First Class with Distinction

Higher Secondary (HSC) and Secondary School (SSC), Science — **2018 and 2020**

BIAM Model School and College, Bogura, Bangladesh · GPA 5.00 / 5.00 in both

SKILLS

Languages	Python, TypeScript, SQL
AI and ML	LLM integration (Anthropic, OpenAI, xAI, Google), RAG, pgvector, prompt engineering and injection defence, machine learning, deep learning, predictive modelling, hyperparameter tuning, anomaly detection, time series (ARIMA, LSTM)
Backend	FastAPI, SQLAlchemy 2, PostgreSQL, Alembic, Redis, OAuth2, REST APIs, Django, Flask
Frontend	React 19, Tailwind, React Query, Vite, Streamlit, UI/UX

Tooling	Docker Compose, GitHub Actions, AWS, Linux, Git, pytest, Vitest, Postman, Paddle billing, AI assisted development (Claude Code)
Data and BI	Data analysis, Tableau, dashboarding, KPI tracking, forecasting, data storytelling, Google Sheets, Google Analytics
Spoken	Bengali (native), English (full professional), Hindi (elementary)

PUBLICATIONS

Machine Learning Classifiers and Hyperparameter Tuning to Enhance Prediction of ASD: An Automated Django Application
2024 2nd World Conference on Communication & Computing (WCONF), IEEE · DOI: 10.1109/WCONF61366.2024.10692172

Predicting Psychological Well-being Using Machine Learning with Hybrid Hyperparameter Tuning Techniques
2024 International Conference on Artificial Intelligence and Emerging Technology (Global AI Summit), IEEE · DOI: 10.1109/GlobalAISummit62156.2024.10947775

A Comparative Study on Hybrid Machine Learning Voting Classifier Models for Alzheimer's Disease Prediction
2023 International Conference on AI for Innovations in Healthcare (ICAIIHI), IEEE · DOI: 10.1109/ICAIIHI57871.2023.10489772

Augmenting Human Capabilities: AI's Impact on Collaborative Workflows

A Roadmap to a Sustainable Future: Empowering Businesses and Consumers Through AI-Driven Green Marketing

PROJECTS

MediGuard AI: drug and side effect predictor with a RAG chat assistant github.com/MediGuardAI
I led this one. A fine-tuned LLM checks a patient's medications for interactions and likely side effects, with a RAG chat assistant on top for follow-up questions. It runs as a Streamlit app.

Multilingual retrieval-augmented generation system github.com/MasrukHabib/Multilingual-Retrieval-Augmented-Generation-RAG-System
Answers a question asked in one language using documents written in another. Embeddings and vector search keep every answer tied back to a real passage instead of invention.

UR3 CobotOps: predictive maintenance for a collaborative robot github.com/MasrukHabib/UR3-CobotOps-Enhancing-Cobot-Efficiency-with-ML
ARIMA, LSTM, K-Means and a few classifiers over the UCI cobot dataset, used to catch anomalies while the robot is running and predict failures before they happen.

ASD Predictor: automated Django application github.com/MasrukHabib/ASD_Predictor_ML_Django_App
A Django app for early autism screening, built on SVM, logistic regression, extra trees and decision trees over UCI data. Tuning brought precision and recall to 1.0. Published at WCONF 2024.

Psychological well-being prediction github.com/MasrukHabib/Psychological_Wellbeingness_Prediction
A web app that scores psychological well-being. I compared logistic regression, decision trees, random forest, gradient boosting and SVM, then tuned them with grid, random and Bayesian search.

Hypothyroidism prediction, real-time Flask app
A research project with collaborators. A Flask app predicts hypothyroidism in real time, with the tuning work needed to keep the results trustworthy.

Phishing website detection github.com/MasrukHabib/Phishing-website-detection-using-machine-learning
A classifier that spots malicious sites from their URL and page features, so someone gets warned before they click.

Tableau dashboards public.tableau.com/app/profile/masruk.habib
Public dashboards covering marketing campaigns, Superstore sales history, and HR analytics and operations.

CERTIFICATIONS

Explore Machine Learning using Python (Infosys Springboard, Sep 2024)
AWS Academy Graduate, Cloud Foundations (Amazon Web Services, Oct 2023)
Database Programming with SQL (Oracle Academy, Nov 2022)
Python Data Structures (Coursera, Aug 2022)
Digital Security Essentials
Student Research Internship Training, SRIT (Marwadi University, Jul 2024)
CCNAv7: Introduction to Networks (Cisco Networking Academy, Feb 2023)
Postman API Fundamentals Student Expert (Postman, Apr 2023)
Programming for Everybody (Coursera, Jul 2022)
Claude Code 101
The Complete Tableau Bootcamp for Aspiring Data Scientists (Infosys Springboard, Mar 2023)
NDG Linux Essentials (Cisco Networking Academy, Apr 2023)
Project Management Foundations (LinkedIn, Jul 2023)
Machine Learning with Python: Foundations

AWARDS

Winner, Azadi Ka Amrit Mahotsav Hackathon 2022 · Government of Gujarat **Mar 2023**
Our team won this state-level hackathon for a project the judges thought could work in practice, not just on a slide.